



UC BERKELEY MIDS · W210 CAPSTONE · SUMMER 2026

Eatelligence

Food Safety Intelligence

A predictive, explainable food-safety risk layer for every licensed establishment in
Chicago, New York City & Los Angeles

Arun Agarwal

Bella Davies

Deepak Srivastava

Aurelia Yang

Jun Xu

Meet the Team



Arun Agarwal

Data Engineering

Data ingestion,
S3 data lake



Bella Davies

Core ML · Evaluation

Modeling, Eval, SHAP,
Fairness Audits.



Deepak Srivastava

**Conversational
Agent**

Agent deployment,
tools, guardrails



Aurelia Yang

**Deep Learning
Experiments**

EDA,
App UX



Jun Xu

Team Lead · PM

Web app,
Deployment

Millions Get Sick Each Year, but Inspections Alone Can't Keep Up

48M

Americans fall ill from
foodborne disease every
year

128K

hospitalizations

\$48B

food-safety
testing by 2033

3,000

deaths

1 in 6

Americans affected
annually

\$1.4 → 3B

compliance
software to 2033

sources:
[CDC Facts About Food Poisoning](#)
[Food Safety Testing Market Report](#)
[Compliance Software Market Report](#)

We Turn Public Data into Actionable Risk Scores

Inspection History

Violation Codes

Business Licenses



**Unified and Explainable
Predictive Risk and Trend Score**

Both Diners and Inspectors Need Better Risk Visibility

CITY RESIDENTS

“
Yep, if it's not an A I'm not going in.

- NYC diner¹

“
Seeing cockroaches is enough for me to never go back.

- Chicago consumer²

“
My wife only eats at A-rated restaurants... if it's not an A she won't go in.

- LA diner

HEALTH INSPECTORS

“
We know that people clean when they know we are coming.

- Health inspector

“
People don't wash their hands... even when they are being inspected.

- Health inspector

“
A restaurant had sewage leaking into their ice machine... they were still serving the ice.

- Health inspector

User Feedback Sharpened Our Focus on Public Health

Primary Customer & Value

“ We need to know who the platform serves. Is it for the health department managers, or for the inspectors? ”

Focus on Public Health

" The inspector's end goal isn't to maximize violations but to promote compliance and protect public health."

Support Jurisdiction Operations

“ Every city is different. A policy that works in Chicago might not fit a suburban village.”

See the Platform in Real Time

We Leverage Public Data for Chicago, NYC, and LA

	Chicago · PRIMARY	New York City	Los Angeles
Sources	Inspection history Violation comments Business licenses	Inspection history	Inspection history Violation history
Variety	69% restaurants 31% food establishments (Grocery, schools, daycares, hospital kitchens, bakeries, caterers, mobile vendors)	Restaurants only	75% restaurants 25% retail food markets
Pipeline	Datasets joined on the business license number	Collapsed raw violation records to inspection events	Datasets joined on the inspection's serial number
Train	01/2019 - 06/2024 (79,268 rows, 77%)	07/2022 - 09/2024 (35,190 rows, 66%)	04/2023 - 07/2024 (36,900 rows, 72%)
Val	07/2024 - 06/2025 (16,564 rows, 16%)	10/2024 - 03/2025 (8,773 rows, 16%)	07/2024 - 12/2024 (7,374 rows, 14%)
Test	07/2025 - 12/2025 (7,008 rows, 7%)	04/2025 - 06/2026 (9,456 rows, 18%)	01/2025 - 03/2026 (7,197 rows, 14%)

All sources: city open-data portals · chronological train / validation / test windows per city, never shuffled

Each City Required a Different Data Strategy

Every city grades differently

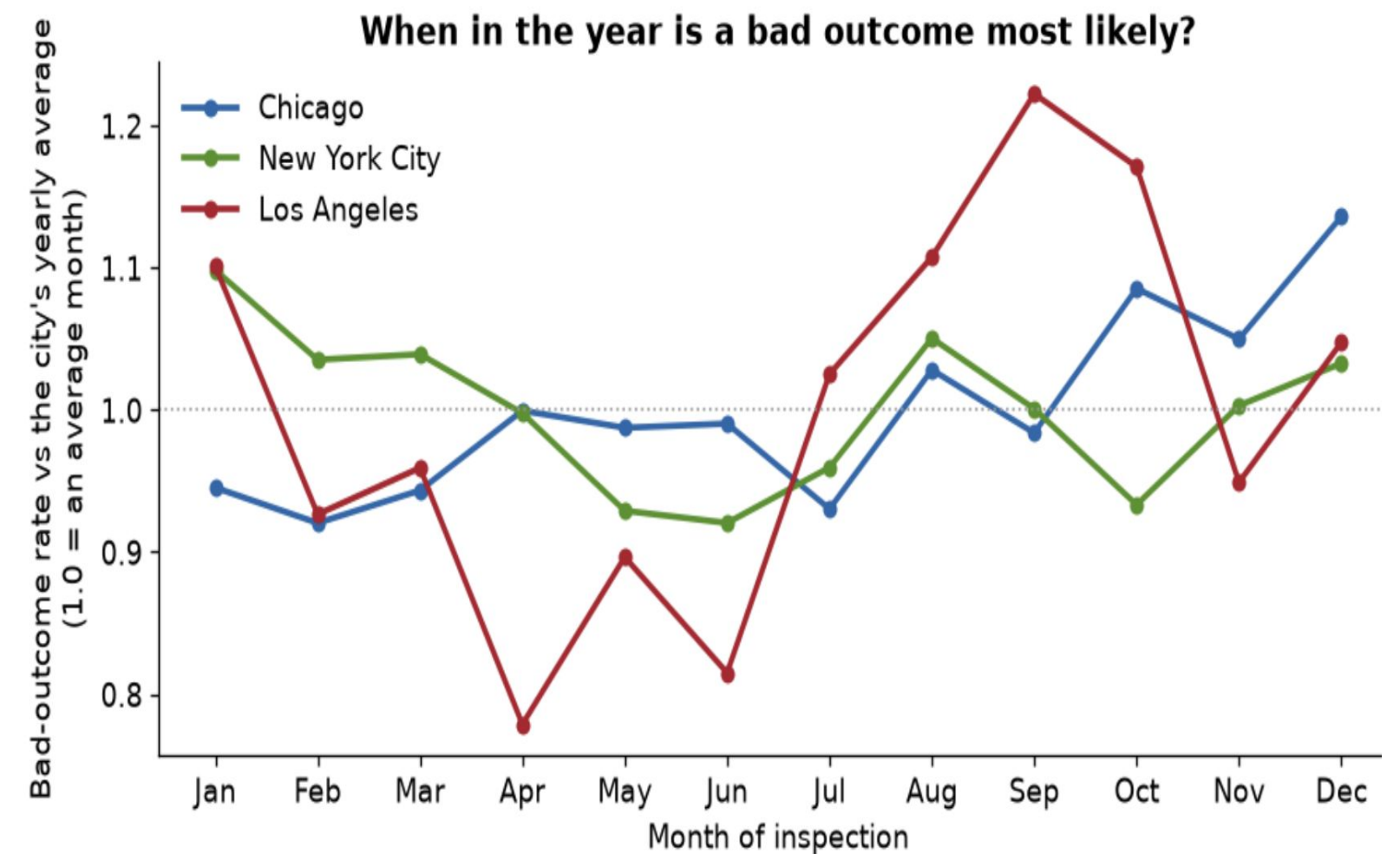
* Prediction target

- **Chicago:** Pass, Pass w/ Conditions, ***Fail w/ violation codes**
- **NYC:** Fewer points is cleaner; A = 0-13, ***B = 14-27, C = >28**
- **LA:** Higher score is cleaner; A = 90-100, ***B = 80-89, C = <70**

Chicago data required data cleaning

- Out of business establishments
- Duplicate license entries from closed-and-reopened venues
- Frequent complaint inspections & re-inspections in Chicago
→ predict bad outcomes occurring in the next 180d

Seasonality Effects



Risk Scores Must Reflect Each City's Reality

Base rates vary dramatically for each city's bad outcome prediction target



Predicted risk scores placed into Risk Tiers, relative to each city's overall performance

City	Bad Outcome Target Variable	Full Dataset Base Rate	Test Dataset *Base Rate* (most recent data)	<50% base rate ● Low	50-100% base rate ● Moderate	100-200% base rate ● Elevated	>200% base rate ● High
Chicago	Fail or Priority violation in the next 180 days	14%	11%	0 - 5%	5 - 11%	11% - 22%	> 22%
NYC	B or C grade next inspection	38%	41%	0 - 20%	20 - 41%	41 - 82%	> 82%
LA	B or C grade next inspection	6%	9%	0 - 4%	4 - 9%	9 - 18%	> 18%

We Improved Chicago's Model While Removing Bias

Original Model

LogReg · 16 features, 2011–2014

We keep its key **feature family**:

- Prior fails, critical violations, trailing violation rate
- Days since last inspection

Not kept:

- Weather features → calendar features
- Socio-demographics, inspector identity, alcohol/tobacco license
- Next routine visit target label

Target Label

We try multiple targets for the outcome predictor variable

Label	PR-AUC	Lift over random
180d	0.344	**3.9x**
365d	0.498	1.68x
Next routine visit	0.604	1.28x

Simple Trees Outperformed More Complex Models

Logistic Regression

BASELINE

XGBoost

SERVED

Neural Networks

EXPLORED

Class Balancing

(scale_pos_weight)

Chicago · 14.5% train prevalence → **5.9×**

NYC · 35.6% train prevalence → **1.8×**

LA · 4.6% train prevalence → **20.6×**

Calibration & Explanation

- Sigmoid grounds probabilities after reweighting
- TreeSHAP explains top drivers of scores

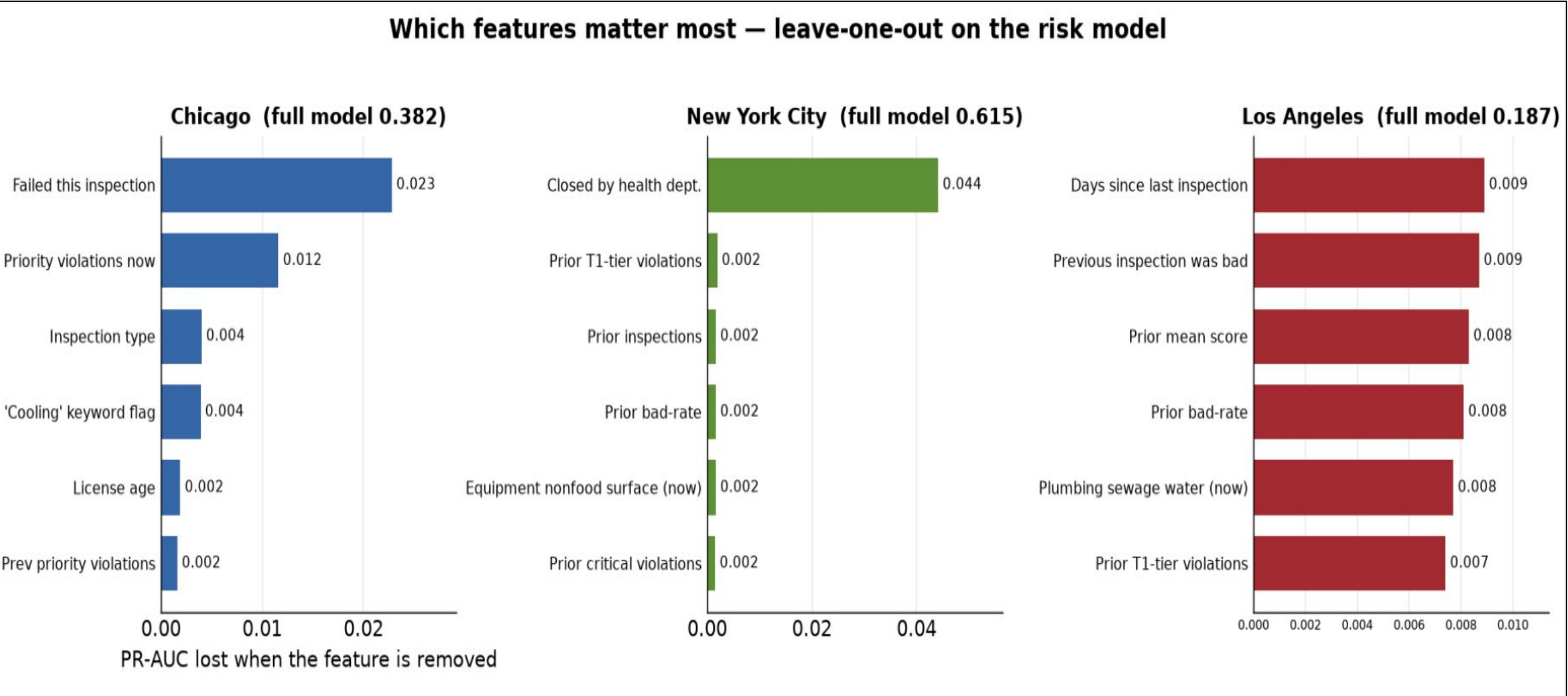
Features Represent Inspection and Violation Patterns

Feature group	Chicago	New York City	Los Angeles
Prior history	8	5	5
Recency and trend	9	6	6
Latest inspection outcome	3	5	4
Latest violation severities + themes	12	14	13
Chicago specific (basic tier, license, complaint/reinspection)	4	—	—
Establishment record (tenure, past closures)	—	2	—
Total features	36	32	28

A second “trend” model

Deliberately blind to the latest visit. The primary risk model is dominated by “what just happened” so it can’t speak to direction.

Inspection History Is the Strongest Predictor of Future Risk



The latest inspection features provide the most information gains to the model.

NYC inspections can result in temporary restaurant closures.

We Unified Three Different Inspection Systems into One Language

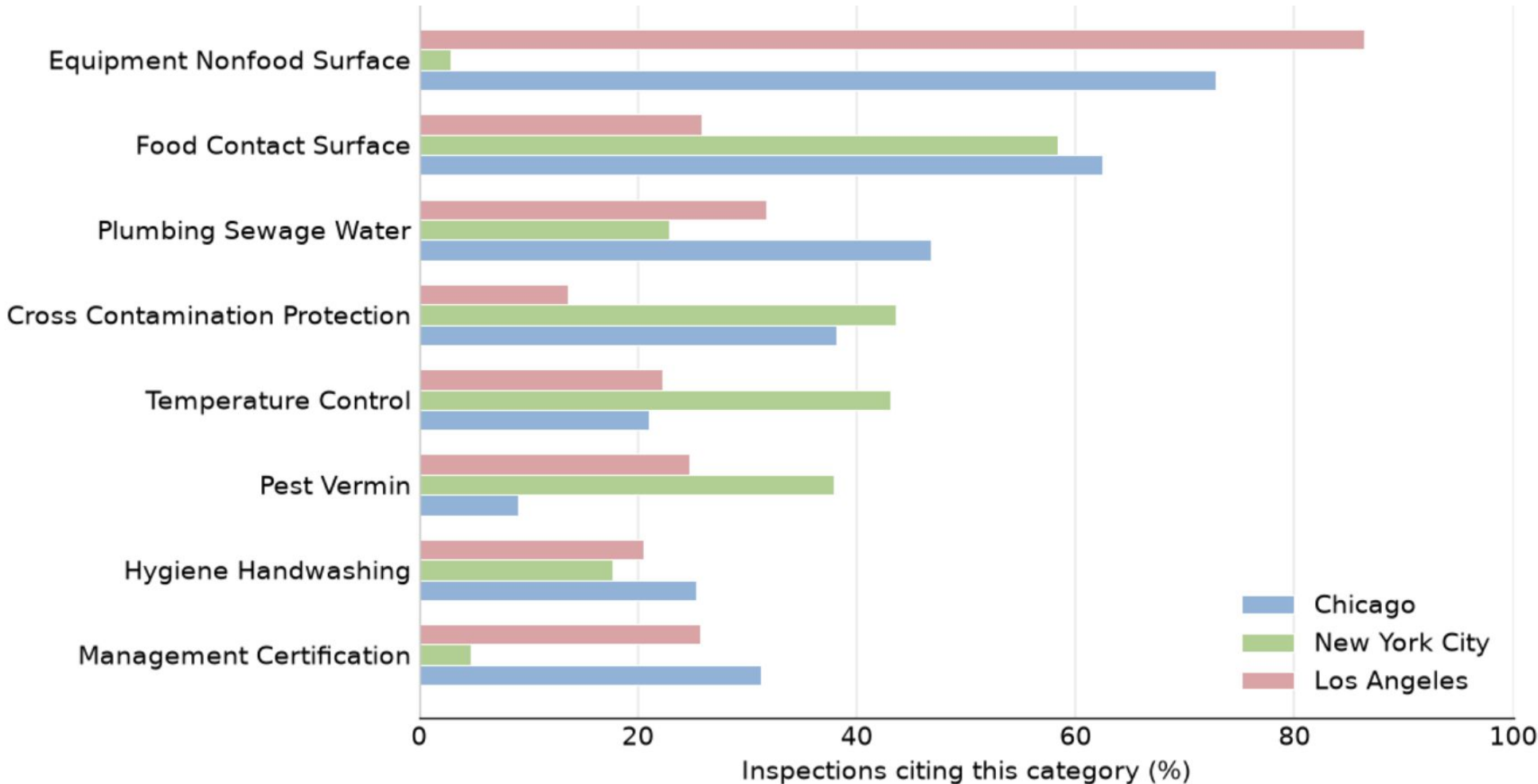
METHOD

Mapping Theme and Severity

- **Theme:** Mapping of 11 categories shared across violation codes in every city.
- **Severity:** from each city's own judgement:
 - Chicago's priority band
 - Priority violation codes 1-29 vs Core violation codes 30+
 - NYC's critical flag & hazard points
 - LA's point weighting

THEMES

% of inspections citing top themes across cities



XGBoost Delivered the Best Performance Across Cities

Final Model Architecture & Performance			
Key Settings (XGBoost)	Chicago	New York City	Los Angeles
Trees / Max Depth	300 / 3	400 / 2	400 / 3
min_child_weight	10	20	5
Served PR-AUC*	0.382	0.615	0.187
Top-decile lift	3.9x	1.8x	2.5x
Key Insight: Signal flips by city. Chicago relies heavily on the latest inspection, while NYC and LA lean more on historical data due to less frequent visits.			

We Evaluated Models Using Metrics That Match Real Inspection Workflows

GATING METRICS

- PR-AUC: because positives are rare
- Precision@10%: the operating point

HEADLINE METRICS

- Lift@10%: interpretable operating point
- ROC-AUC: translatable prediction performance

Metric	Chicago	NYC	LA
PR-AUC	0.38	0.62	0.19
Precision@10%	0.42	0.74	0.22
Lift@10%	*3.9x*	1.8x	2.5x
ROC-AUC	0.81	0.70	0.72

The Model Finds High-Risk Restaurants and Clearly Communicates Its Limits

Chicago predictions are strongest

A ranked worklist for inspectors is the best use case: the top 10% catches up to 3.9× what random inspection order finds.

City	Base rate	Precision @10%	Lift
Chicago	11%	0.42	3.9×
NYC	41%	0.74	1.8×
LA	9%	0.22	2.5×

Chicago Nearly blind to brand-new restaurants: on venues that have never failed, PR-AUC drops from 0.38 to **0.13**.

NYC The “Low” tier still observes **23% actual failures**. UI says “lower risk for NYC,” never “safe.”

LA False positive rate varies **0.14 across ZIPs**, despite ZIP being dropped.

Every Risk Prediction Can Be Explained

		← LOWERS RISK	RAISES RISK →
	01 Failed the current inspection ⓘ Whether this establishment failed its current inspection: the strongest near-term signal of forward risk.		+1.17
	02 6 priority violations at this inspection ⓘ Priority (code 1–29) violations cited at the current inspection: the serious, higher-risk hazards.		+1.09
	03 12 core violations at this inspection ⓘ Core (lower-tier) violations at the current inspection: minor issues that still add up.		+0.37
	04 Improper cooling cited Improper cooling of food cited in recent violations: a common cause of bacterial growth.		+0.26
	05 8 prior complaint-driven inspections How many past inspections were triggered by a complaint rather than routine schedule.		+0.21

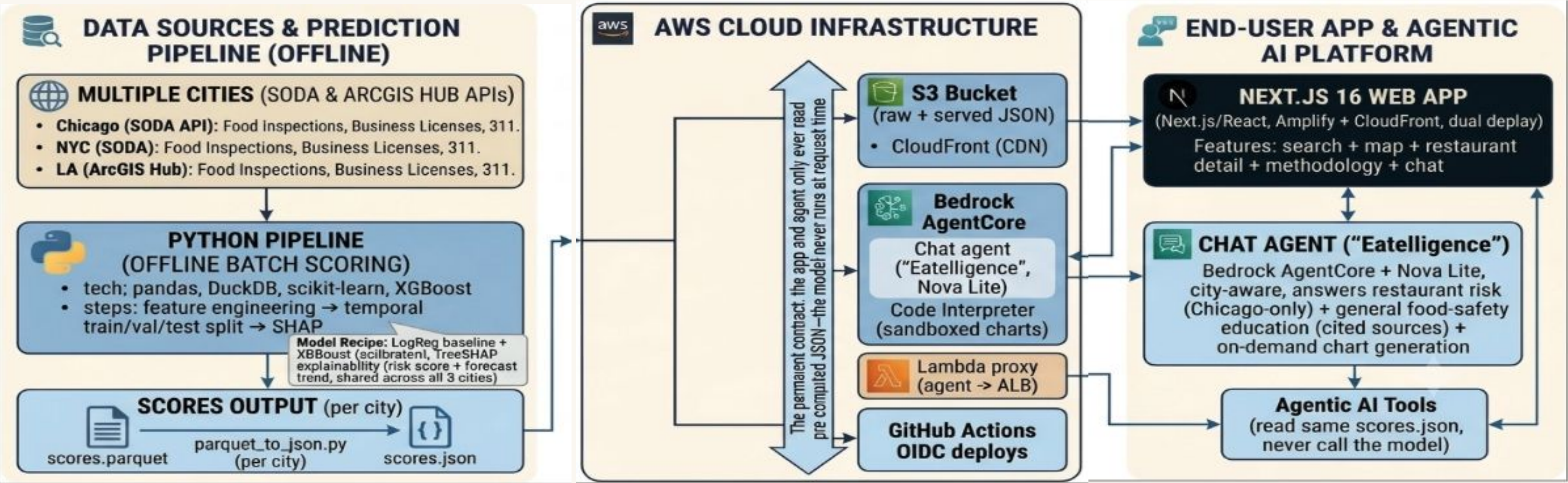
A Production Pipeline Powers Predictions, Explanations, and the App

THREE KEY TAKEAWAYS

1. **Batch-scored, not real-time:** predictions computed offline and served as static JSON via S3/CloudFront — keeps app fast (<5s load), cheap, and model auditable, but scores refresh on a schedule.

2. **Multi-city, one pipeline:** same XGBoost + SHAP architecture now scores Chicago, NYC, and LA from different raw data sources, reconciled through a shared severity-tier crosswalk — approach generalizes.

3. **Rigor over raw performance:** all changes evaluated by leak-free temporal split, promo gate, and formal fairness audit — rejected promising wins (e.g. deep learning models, alcohol/tobacco licenses) when they didn't hold up under honest evaluation.



Our AI Assistant Answers Food Safety Questions Responsibly

AI Assistance Tools: Task-Specific for Active City



find_inspection_records

explain_restaurant

optional:

food_safety_info

find_reviews

visualize_data

- Strands agent on Amazon Nova 2 Lite via Bedrock
- Reads the same published scores as the app

Agent Framing:

- General pages
- Establishment page
- Inspector page
- Caregiver page

Guardrails:

- “Risk signal” framing (no verdict)
- Active city + general food-safety education only
- No numbers without a tool result
- Cite returned source
- No personalized advice

We Evaluate Accuracy, Fairness, and Safety as well as Chat Quality

Deterministic

- ✓ Faithfulness
- ✓ Name lookup
- ✓ Citations

LLM judge

- ✓ Tone
- ✓ Fairness
- ✓ Boundaries

The Winning Formula: Better Features, Boosted Trees, and Explainability

- More diverse datasets $\neq$ better signal
- Trees beat fancier architectures
- Explainability and trustworthiness

We're Ready to Scale Beyond Three Cities

- Automated weekly refresh
- Inspector accounts & chat history
- Onboard more cities and demo to city health depts for powering inspector worklists

EATELLIGENCE

**Build explainable food-safety intelligence to
help diners, inspectors, and communities
navigate food safety risk with confidence.**

Thank you · Questions welcome

Acknowledgements & Resources

Acknowledgements

Our instructors and advisors, for weekly feedback that shaped the target, the evaluation, and this presentation · responsible-AI review of trust, privacy, and governance practices · the friends and target users who tested the app and the agent.

Related work & attributions

Original Chicago DPH food-inspection forecasting model (2015) and its 2019 hindsight audit · FINDER (Google search-location) paper · CDC foodborne-illness estimates · Chicago, NYC & LA open-data portals.

APPENDIX

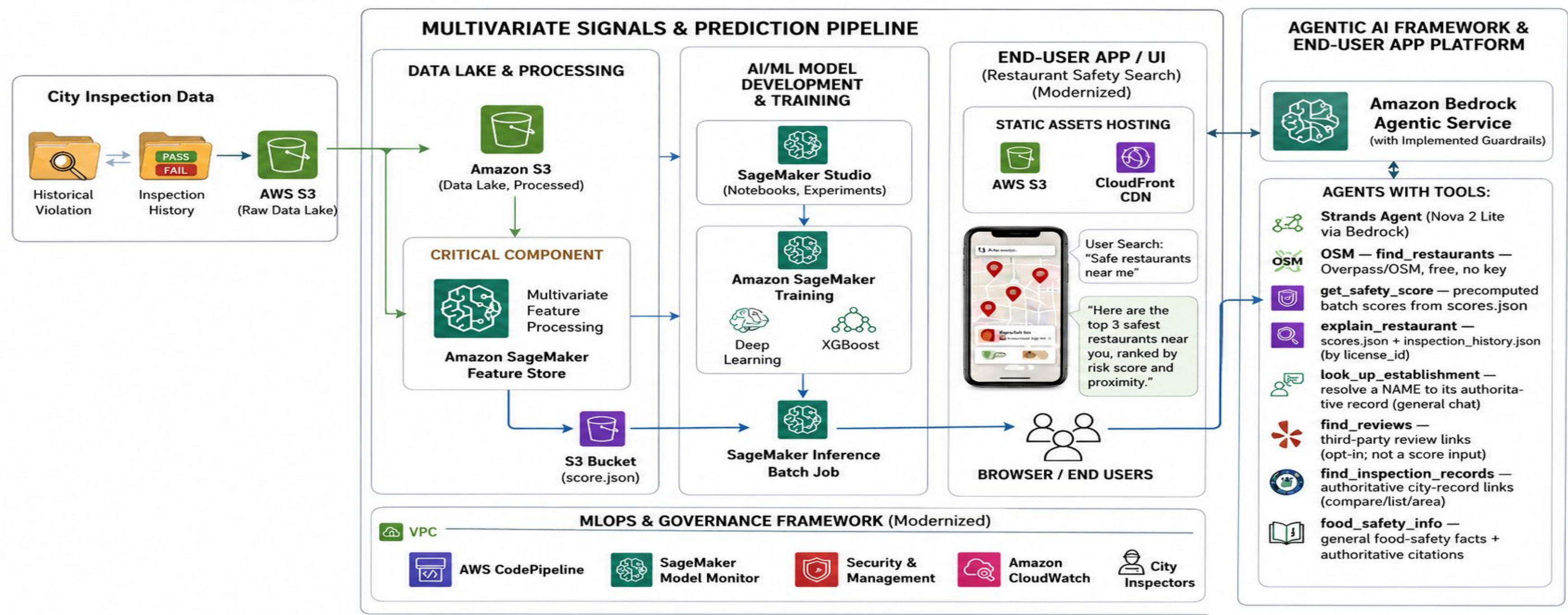
Team Contributions

Member	Contributions
Jun Xu	Team lead · web app & UX · AWS deployment (CloudFront, Amplify) · multi-city product
Arun Agarwal	Data engineering · SODA ingestion · S3 data lake · refresh pipeline
Bella Davies	Feature engineering · XGBoost + calibration · evaluation framework · fairness audits · SHAP serving
Deepak Srivastava	Chat agent (Bedrock / AgentCore) · guardrails · agent evals · feature ablations
Aurelia Yang	EDA & label construction · deep-learning experiments (MLP, HPO) · app UI/UX review

System Architecture

Key Takeaways

- Real-time ingestion scaled back; focus on high-signal City Inspection Data.
- Pre-computed risk scores served via a cost-efficient batch inference process.
- **Agentic AI** assistant provides context-aware, guardrailed search experience.



Our AI Assistant Answers Food Safety Questions Responsibly

AI Assistance Tools: Task-Specific for Active City

find_restaurants → look_up_establishment → get_safety_score

find_inspection_records

explain_restaurant

optional:

food_safety_info

find_reviews

visualize_data

- Strands agent on Amazon Nova 2 Lite via Bedrock
- Reads the same published scores as the app

Framing: scoped contexts within the agent

- **General pages:** scoped to the active city
- **Establishment page:** scoped to the specific establishment page
- **Inspector page:** framed for violation history, compliance drivers, and worklist triage.
- **Caregiver page:** framed for the critically ill, immunocompromised, elderly, or children,

Guardrails: prompt + Bedrock Guardrails

- “Risk signal” framing: not a safe / unsafe verdict
- Active city + general food-safety education only
- No numbers without a tool result
- Cite the returned source for any general fact
- No personalized medical or legal advice

A Production Pipeline Powers Predictions, Explanations, and the App

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